Data Mining & Machine Learning
M.Sc. in Artificial Intelligence and Data Engineering
University of Pisa

Project Report

Academic Year 2025–2026

Project Title: Diabetes Check: A Multiclass Machine Learning Approach for Early Clinical Screening

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Submission date: 12 June 2026

Abstract

This project addresses the public health challenge of diabetes risk assessment by leveraging the BRFSS 2015 epidemiological dataset (253,680 records, 21 predictive features) for a multi-class classification task (healthy, prediabetic, diabetic). To counter severe class imbalance, a robust pipeline integrated exploratory data analysis, outlier detection, and stratified group partitioning. Five machine learning algorithms were optimized via GridSearchCV and rigorously evaluated using a Nested Stratified Group k-fold Cross-Validation framework, with statistical significance validated through the Wilcoxon signed-rank test.

Explainable AI (SHAP) was integrated for global and local interpretability. Empirical results showed that cost-sensitive Logistic Regression and XGBoost achieved comparable top-tier performance under the balanced accuracy metric. Logistic Regression was selected as the optimal clinical model due to its lower architectural complexity, with self-reported general health, high blood pressure, and body mass index emerging as the primary risk predictors. Although not suitable for autonomous deployment due to constraints in capturing atypical phenotypes, the optimized framework delivers a transparent framework for early patient screening, serving as a valuable decision-support tool for preventative healthcare.

1. Introduction

Diabetes mellitus represents a critical and expanding global healthcare challenge due to its progressive and frequently asymptomatic beginning. This study addresses the multi-class risk classification problem distinguishing between healthy, pre-diabetic, and diabetic individuals to provide an interpretable decision-support framework capable of guiding targeted preventive interventions and optimizing public healthcare expenditures.
While previous literature predominantly focuses on binary classification (Xie et al., 2019 [1]) or standard three-class architectures without addressing severe class imbalances (Cekay, 2024 [2]), this project explicitly preserves the clinical granularity of the pre-diabetic state.
By leveraging advanced machine learning paradigms and rigorous optimization, the study aims to extract robust predictive patterns from large-scale epidemiological data without replicating simplified baseline pipelines.
The project is structured around three core objectives:

1. Predictive Modeling: Developing a multi-class classifier using heterogeneous socio-demographic and behavioral health indicators
2. Rigorous Validation: Benchmarking diverse algorithmic families (tree-based ensembles, probabilistic, and linear models) via a robust Nested Stratified Group k-Fold Cross-Validation framework
3. Clinical Interpretability: Integrating Explainable AI (XAI) via SHAP values to deliver global and local transparency, ensuring the model's utility as a trustworthy screening tool

References:

- [1] Xie Z, Nikolayeva O, Luo J, Li D. Building Risk Prediction Models for Type 2 Diabetes Using Machine Learning Techniques (2019, CDC)
- [2] Cekay N. Diabetes Predictions using Classification Models (2024, Kaggle)

2. Problem

We classify patients' diabetic status using a multi-class approach. The target variable *Diabetes_012* consists of three classes: no diabetes, pre-diabetes, and diabetes. Input features come from the CDC 2015 BRFSS phone survey and represent self-reported health factors. The dataset includes socio-demographic variables, behavioral habits, and clinical indicators. Table 1 summarizes the problem specifications.

Table 1. Problem and dataset at a glance

Attribute	Summary
Task	Multiclass Classification (3 classes)
Dataset	CDC BRFSS 2015 — 253,680 samples, 22 features
Target variable	Diabetes_012 (0=Healthy, 1=Pre-Diabetic, 2=Diabetic)
Features	21 input features + 1 target variable (22 columns total)
Missing values	0% missing values
Class distribution	84.2% / 1.8% / 13.9% — highly imbalanced
Evaluation metrics	Balanced Accuracy (primary), F1-macro, Precision- Recall
Split	Nested Stratified Group 10-Fold CV (inner: 5-Fold)

2.1 Exploratory Data Analysis

The dataset contains no missing values across features or classes, confirming full data integrity. The target variable shows a severe class imbalance. Healthy individuals (Class 0) represent 84.2% of the samples, diabetics (Class 2) represent 13.9%, and pre-diabetics (Class 1) account for only 1.8% (Figure 1).

Spearman correlation coefficients show no collinearity. No feature pair exceeds the $|r| > 0.7$ threshold, indicating absence of redundant variables. General health perception (`GenHlth`) has the highest positive correlation with the target ($r = 0.30$), followed by high blood pressure (`HighBP`, $r = 0.27$) and body mass index (`BMI`, $r = 0.24$). Socio-economic variables like income ($r = -0.17$), education ($r = -0.13$), and lifestyle behaviors like physical activity ($r = -0.12$) show negative correlations (Figure 2).

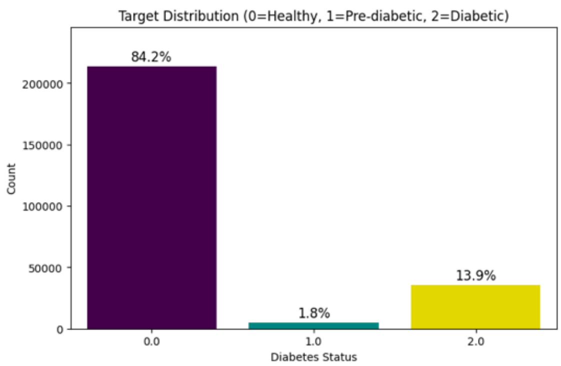

Figure 1. Target variable distribution showing class proportions

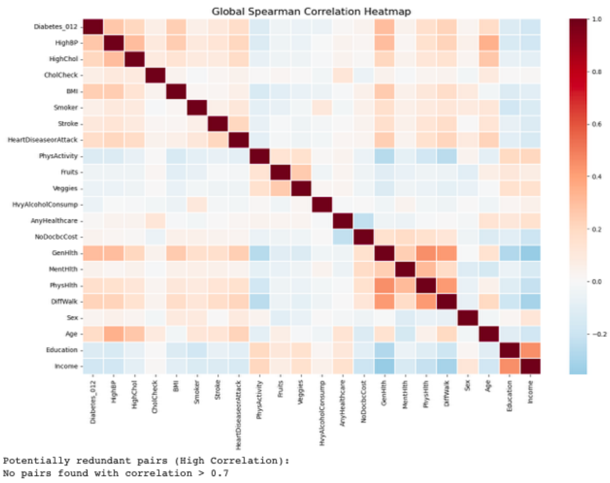

Figure 2. Global Spearman correlation heatmap for the selected features

2.2 Feature Description

The set of features includes binary, ordinal and continuous variables chosen for their clinical and epidemiological relevance to diabetes pathogenesis. We include demographic (age, sex) and socio-economic factors (income, education) to capture the biological and social context of each patient. We integrate behavioral lifestyle indicators (smoking, physical activity, fruit/vegetable consumption) and pre-existing medical conditions (high blood pressure, high cholesterol) as direct risk factors. Features with individually low marginal correlation, such as AnyHealthcare, demonstrate significant discriminatory value through non-linear interactions with socio-economic variables (e.g., Income), justifying their inclusion in the model.

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3. Pipeline and Experiments

3.1 Preprocessing

We apply no data imputation since the dataset contains no missing values. We omit categorical encoding because all variables already possess a structured numerical format. We retain outliers in `BMI` due to their clinical relevance in diabetes pathogenesis.
To strictly prevent data leakage, we perform feature scaling using StandardScaler within a machine learning Pipeline. This approach learns the scaling parameters exclusively on the training set of each cross-validation fold. Finally, we generate a profile_id by grouping features to handle duplicate and ambiguous profiles. We utilize this identifier as the grouping variable within a StratifiedGroupKFold split. Consequently, we ensure that identical profiles always remain within the same fold, preventing data leakage and guaranteeing a reliable evaluation of model generalization.

3.2 Feature Selection and Engineering

We retain all 21 original predictive features and skip automated feature selection. The Spearman correlation matrix justifies this choice (Figure 2).
We also avoid dropping variables with near-zero linear correlation with the target. For instance, *AnyHealthcare* exhibits a baseline univariate correlation of just 0.01. However, bivariate interaction analysis with `Income` reveals a significant diagnostic bias (Figure 3). At low-income levels (*Income* 2.0), insured individuals show a 27.2% diabetes prevalence compared to 15.7% for uninsured individuals. This 11.5% gap demonstrates how the variable captures crucial structural nuances in data reporting, justifying its retention.

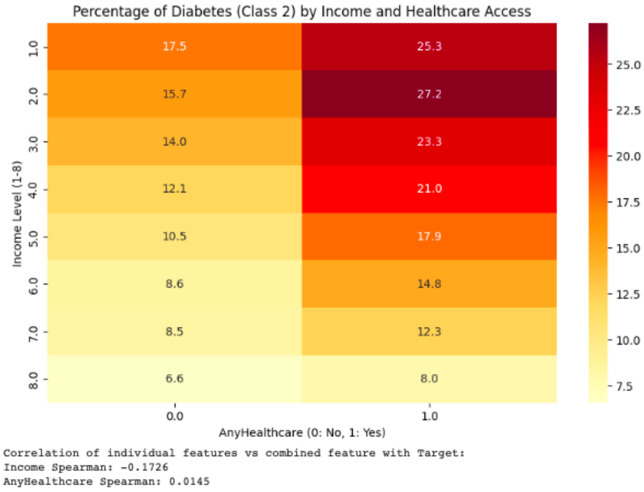

Figure 3. Diabetes percentage (Class 2) by income level and healthcare access

3.3 Model Selection and Cross-Validation

We evaluate five distinct algorithmic families: tree-based ensembles (Random Forest, AdaBoost, XGBoost), a probabilistic model (Gaussian Naive Bayes), and a linear model (Logistic Regression). To guarantee unbiased evaluation and prevent data leakage, we implement a Nested Stratified Group 10-Fold Cross-Validation. The inner loop (5-fold) optimizes hyperparameters via GridSearchCV. The outer loop (10-fold) assesses generalization by partitioning data through profile_id. This grouping variable ensures that identical patient profiles are isolated within the same fold, forcing the models to evaluate on unseen patient structures rather than memorized duplicates. We fit all transformations strictly within the training folds. We handle the severe class imbalance using cost-sensitive configurations: class_weight='balanced' for Random Forest, AdaBoost, and Logistic Regression; compute_sample_weight for XGBoost; and uniform class priors for Gaussian Naive Bayes. Models are ranked via Macro F1-score and Balanced Accuracy (Table 2). We assess candidate models using Figure 4 to evaluate cross-validated performance stability and Figure 7 to analyze the macro-averaged Precision-Recall operating points. This joint evaluation justifies selecting Logistic Regression as our final architecture, balancing high recall with competitive precision and optimal variance across splits.

Table 2. Comparison of candidate models.

Model	Hyperparameter tuning	CV Strategy	Val. Metric	Test Metric
Random Forest	Grid Search	Nested Stratified Group 10-Fold CV	0.5118 ± 0.0012	0.5125 ± 0.0052
AdaBoost	Grid Search	Nested Stratified Group 10-Fold CV	0.4712 ± 0.0036	0.4671 ± 0.0070
XGBoost	Grid Search	Nested Stratified Group 10-Fold CV	0.5240 ± 0.0017	0.5259 ± 0.0075
Gaussian Naive Bayes	Grid Search	Nested Stratified Group 10-Fold CV	0.4889 ± 0.0007	0.4889 ± 0.0057
Logistic Regression	Grid Search	Nested Stratified Group 10-Fold CV	0.5194 ± 0.0011	0.5200 ± 0.0054

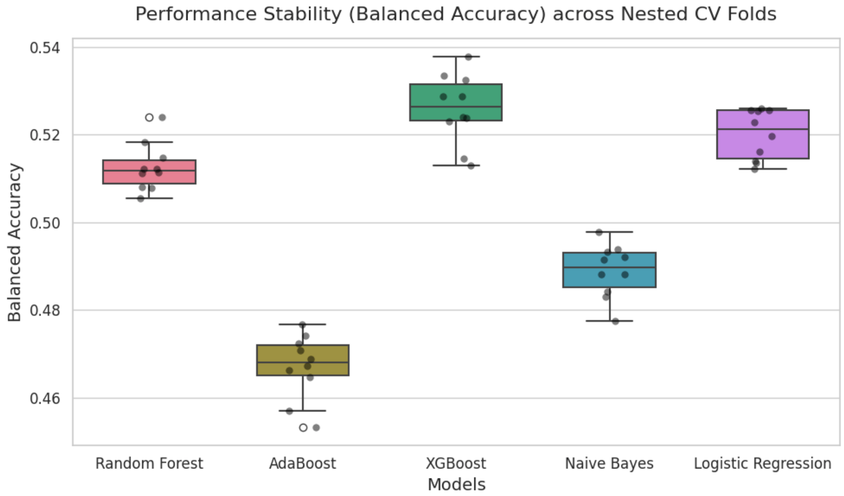

Figure 4. Balanced Accuracy distribution across the 10 outer folds of the Nested CV for all models

3.4 Hyperparameter Optimization

We apply Grid Search across all five models, optimizing Balanced Accuracy within the 5-fold inner loop. Search spaces cover ensemble size, tree depth, and learning rate for tree-based models; var_smoothing for Gaussian Naive Bayes; and regularization strength C for Logistic Regression. Following model selection, we retune the final Logistic Regression on the full dataset over an expanded grid C ∈ {0.001, 0.01, 0.1, 1, 10, 100}. The validation curve (Figure 5) confirms low sensitivity to regularization, with C = 0.1 selected as the optimal configuration.

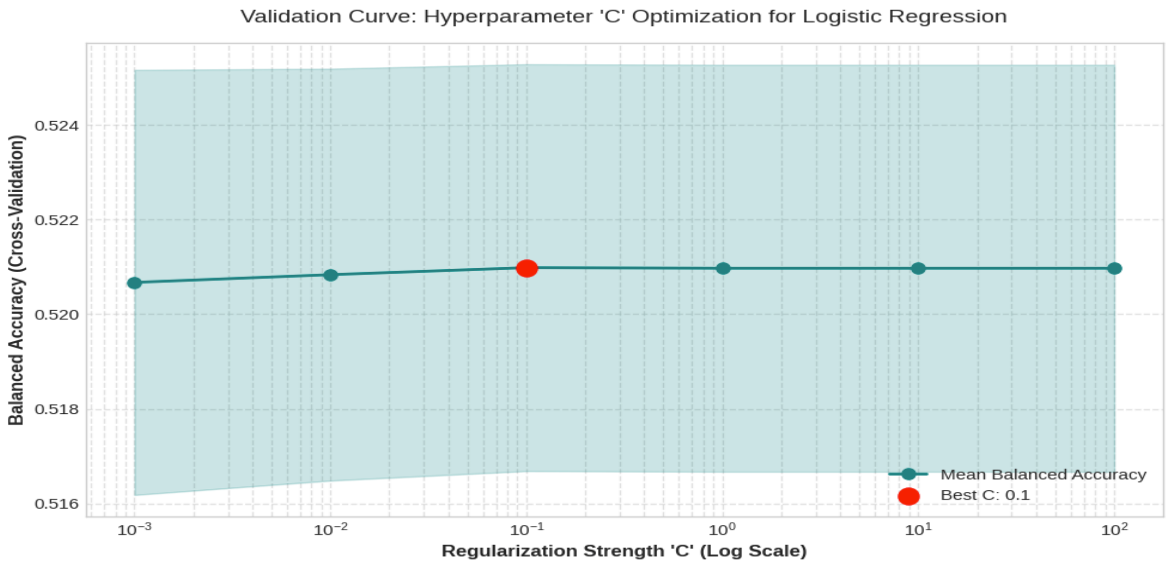

Figure 5. Validation curve for hyperparameter C of the final Logistic Regression

3.5 Statistical Tests

We apply the Wilcoxon Signed-Rank Test on paired Balanced Accuracy scores across the 10 outer CV folds. Logistic Regression significantly outperforms Random Forest (W = 0.000, p = 0.002), while no statistically significant difference emerges against XGBoost (W = 9.000, p = 0.065). These results confirm Logistic Regression as a robust choice, matching the best ensemble while retaining interpretability.

3.6 Final Model Training

Following model selection, we retrain Logistic Regression on the full dataset using GridSearchCV with a 5-fold Stratified Group CV, yielding C = 0.1 as the optimal regularization strength. Generalization performance is not assessed on a separate held-out set; it is instead estimated by the outer loop of the preceding nested cross-validation, which provides an unbiased evaluation on unseen profile structures across all 10 folds.

4. Results and Discussion

4.1 Performance Evaluation

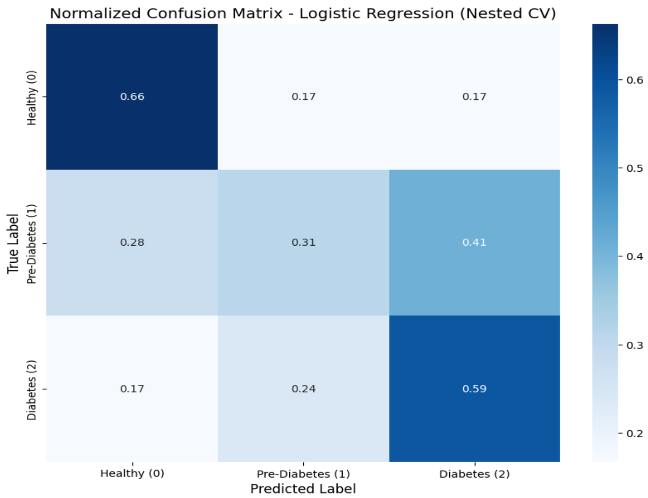

Figure 6. Confusion Matrix Logistic Regression

The model correctly classifies 66% of healthy subjects (Class 0) but misclassifies 34% of them, split equally (17% and 17%) between pre-diabetes and diabetes. Class 1 (pre-diabetes) represents the primary bottleneck: only 31% of these patients are correctly identified. The model misclassifies them as diabetic in 41% of cases and as healthy in 28%, confirming that the intermediate clinical profile is nearly indistinguishable from the other categories under a linear formulation. On Class 2 (Diabetes), the model achieves a 59% recall, though a 24% misclassification toward pre-diabetes reflects blurred clinical boundaries between the two pathological states.

In practice, these errors carry asymmetric real-world consequences. The 34% false-positive rate (Type I error) among healthy individuals induces manageable clinical overhead and temporary patient anxiety, easily resolved via standard secondary testing. Conversely, false negatives (Type II error) pose a severe clinical threat: misclassifying 17% of diabetics and 28% of pre-diabetics as healthy deprives at-risk individuals of timely preventive interventions, accelerating chronic complications. However, by leveraging cost-sensitive weights to prioritize sensitivity over pathological classes, the error distribution aligns with the operational mandate: serving as a safe digital triage tool that prioritizes patient safety over absolute precision. To further validate the model selection, the macro-averaged Precision-Recall operating points of all candidate architectures are compared in Figure 7.

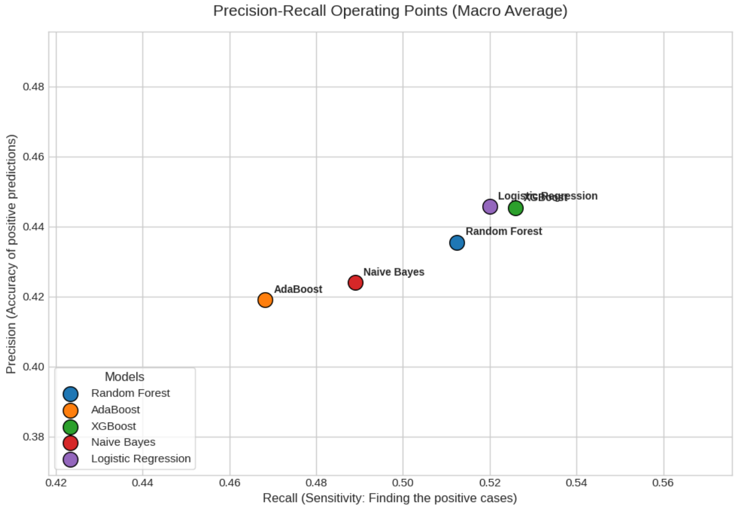

Figure 7. Precision-Recall Comparison

As illustrated in Figure 7, both Logistic Regression and XGBoost establish the most optimal operational trade-offs compared to the other candidate models, occupying nearly identical spaces at the upper-right frontier of the plot. While XGBoost shows a marginal gain in macro-recall, Logistic Regression maintains a slightly higher macro-precision. This visual proximity corroborates the statistical tie previously detailed by the Wilcoxon signed-rank test. Grounded in this empirical equivalence and guided by the principle of parsimony, Logistic Regression was selected as the champion model. Due to its lower architectural complexity and intrinsic clinical transparency, it offers the most efficient and dependable decision-support framework for non-invasive population screening.

4.2 Model Explainability

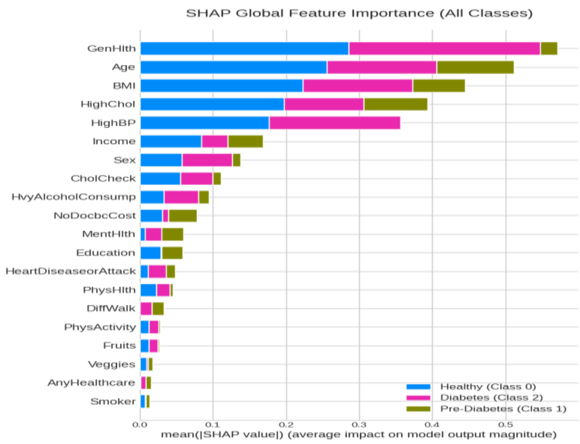

Figure 8. SHAP Global Feature Importance

As shown in Figure 8, global decisions are heavily driven by key clinical risk factors: **GenHlth**, **Age, BMI, HighChol,** and **HighBP**. The high magnitude of SHAP values for the diabetes class (Class 2, magenta) on these variables directly align with established epidemiological literature. Crucially, SHAP contributions for pre-diabetes (Class 1, green) are consistently compressed across all features, proving that the intermediate state lacks a distinct behavioral signature in this survey data. This algebraic compression perfectly supports the low classification sensitivity discussed in Section 4.1. To bridge this global behavior with patient-specific profiles, Figure 9 provides a local waterfall plot evaluating an individual instance.

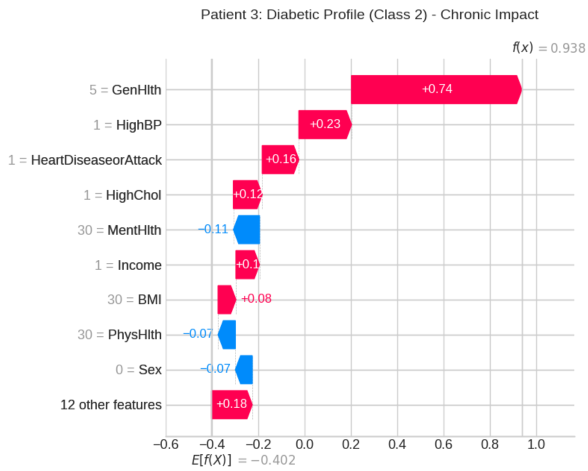

Figure 9. Local SHAP Explanation for a Representative Diabetic Patient (Class 2).

As displayed in Figure 9, the local prediction aligns seamlessly with domain knowledge and reflects the macro-priorities captured in the global analysis. The model's confident decision to classify the patient as diabetic is driven by a severely altered perceived health status (**GenHlth = 5**), which acts as the dominant risk driver. This chronic scenario is systematically reinforced by the core clinical comorbidities identified globally, such as hypertension (**HighBP = 1**), previous cardiovascular events **(HeartDiseaseorAttack = 1**), and hypercholesterolemia (**HighChol = 1**). This demonstrates that for well-defined, traditional clinical phenotypes, the cost-sensitive Logistic Regression operates with high transparency and clinical consistency, successfully aggregating linear risk factors to cross the decision threshold.

4.3 Error Analysis

A deeper investigation into misclassified samples highlights predictable failure modes tied to the linear constraints of the feature space, particularly regarding false negatives. Figure 10 illustrates this critical diagnostic threat, evaluating an atypical diabetic profile (Patient 4) misclassified as healthy.

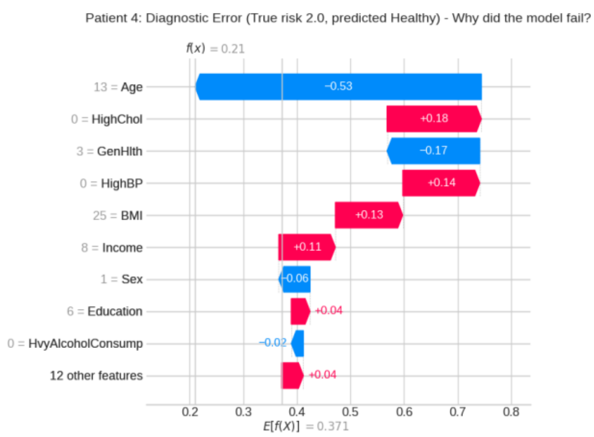

Figure 10. Local SHAP Explanation of a False Negative Error for an Atypical Diabetic Profile

As shown in Figure 10, the error is an explicit product of Logistic Regression's additive nature when handling atypical clinical phenotypes. In this specific patient, the algebraic summation of protective factors, namely the absence of hypercholesterolemia (HighChol = 0) and hypertension (HighBP = 0), paired with a non-pathological BMI of 25 generates a dominant positive force toward the healthy class. This additive effect completely overrides and cancels the opposing biological risk indicator of advanced age (Age = 13), which was trying to push the model away from the healthy prediction. Because a linear framework cannot map complex feature interactions or capture late-onset diabetes in asymptomatic individuals without biochemical data, it is systematically misled by these combinations.

Concurrently, Pre-Diabetes (Class 1) is routinely misclassified due to severe phenotypic overlap. Since self-reported behavioral indicators reflect identical symptoms across both pathological states, the intermediate category lacks a unique algebraic signature, causing high dispersion rates observed in the confusion matrix.

5. Conclusions

This project successfully developed an interpretable, multi-class framework that transitions preliminary diabetes screening from a rigid binary switch into a granular risk-stratification tool, meeting all core research objectives. The empirical findings validate that cost-sensitive Logistic Regression offers an optimal operational profile for population-level triage, delivering a high macro-recall that minimizes missed diagnoses without sacrificing deployment viability. Furthermore, post-hoc explainability via SHAP successfully grounded the framework's behavior within accepted epidemiological literature, ensuring that predictions remain clinically transparent and aligned with established medical domain knowledge.

However, these results represent a diagnostic foundation rather than a standalone clinical solution due to inherent data constraints. The severe class imbalance within the self-reported dataset penalized the classification of the intermediate pre-diabetes category, which lacks a distinct behavioral signature and suffers from massive phenotypic overlap. Additionally, since the cohort is strictly tied to the US population, its international generalization remains unverified. Lastly, the linear boundaries of Logistic Regression create systematic blind spots when handling atypical phenotypes, as the architecture cannot inherently map non-linear feature interactions without explicit biochemical indicators.

Moving forward, future work will focus on scaling this framework into a viable clinical asset. Key priorities include validating the pipeline on international cohorts to ensure multi-ethnic robustness and integrating objective bio-clinical markers (such as HbA1c, family medical history, and waist-to-hip ratios) to supplement self-reported data. Finally, deploying the framework in a primary care clinical pilot will allow for the empirical evaluation of its triage speed and physician trust during early preventive public health interventions.

NOTE: The code and all documentation of the project is available at this link
https://github.com/LucaMinuti/DMML